

UJWAL S R

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Experience

A.P. Moller–Maersk

Bengaluru, India

AI/ML Intern

Sep 2025 – Present

- Architected and built an end-to-end distributed entity resolution system using semantic retrieval, ML-based matching, and a fine-tuned cross-encoder trained on augmented data and processed across distributed GPUs in Databricks, **improving resolution accuracy from 30% to 75%**.
- Optimized Qdrant vector storage to **reduce memory requirements by 95% while retaining 97% precision**, and built controlled PySpark LLM pipelines for large-scale quality validation and batch inference.
- Built Azure Data Factory pipelines to load **50M+ row data** tables from Data Lake into PostgreSQL, implementing indexing strategies and complex data-type handling while **reducing data-loading time by 50%**.
- Developed a graph traversal algorithm in PySpark GraphFrames for efficient connected components and contributed to an agentic AI system orchestrating database access through tool-based agent workflows.
- Developed geographic address standardization and treatment pipelines across 300K+ addresses to improve consistency and usability of location data.

7thGear AI

Bengaluru, India

AI/ML Intern

May 2025 – Aug 2025

- Designed multimodal video processing pipelines combining **computer vision, Vision Transformers, NLP, and LLMs** to extract and structure business relevant information from multimedia data.
- Created data augmentation pipelines and trained custom CNN models for image classification, improving model **accuracy from 60% to 95%**.
- Quantized and Fine-tuned the Llama 3.2 3B model using QLoRA and synthetic data generation to address domain specific requirements and improve task-specific performance.

LimitLessMind.ai

Delaware, USA(Remote)

AI Developer

Jan 2025 – Mar 2025

- Designed data ingestion pipelines to transform **unstructured documents into a GraphRAG** knowledge base containing structured product information, and developed tools for RAG agents to query the graph and answer customer questions.
- Contributed to a peer-to-peer, **session based Voice AI platform using LiveKit** and GPT-Realtime, enabling fast, low latency voice AI for answering customer queries.

Projects

Disaster Relief RL Environment for Multi-Agent RL Training (Meta RL Hackathon)

- Designed an OpenEnv-based disaster relief environment simulating triage, resource allocation, dynamic replanning, terrain constraints, communication failures, and time-critical operations for **multi-agent RL training**.
- Designed a reward system and multi-metric grading** process to evaluate agent decisions for covering task completion, response time, correct resource use, and error handling while preventing agents from reward hacking.
- Deployed the end-to-end environment as a Dockerized FastAPI service with standardized reset, step, grade, and state openENV APIs for integration with RL training workflows.

LLM Transformer Architecture from Scratch

- Implemented a decoder only Transformer language model from **scratch in PyTorch** using the Tiny Shakespeare dataset, building **BPE tokenization, RoPE embeddings, RMSNorm, feed-forward networks**, and end-to-end model components.
- Implemented and evaluated Multi-Head Attention (**MHA**), Grouped-Query Attention (**GQA**), and Multi-Head Latent Attention (**MLA**), comparing accuracy and performance metrics and using MLA in the final architecture.

Agentic AI Self-Healing CI/CD Pipeline

- Developed an agentic AI-powered CI/CD pipeline with build, analysis, fix, and test agents to automatically detect, analyze, fix, and validate software vulnerabilities, using a **fine-tuned Qwen-2.5-Coder** model for test case generation, all executed within a sandboxed CI/CD environment.
- Integrated MCP with a **queryable GraphDB for structured codebase** access, alongside GitHub webhook automation, GitHub Actions, WebSocket-based monitoring, and automated testing for end-to-end self-healing workflows.

Technical Skills

Languages: Python, C++, Java, SQL

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, Computer Vision, Vision Transformers, NLP, Reinforcement Learning

LLM / GenAI: LLM Fine-Tuning (DoRA, QLoRA), Quantization, RAG, GraphRAG, Agentic AI, Multi-Agent Systems, Voice AI (LiveKit, GPT-Realtime), LangChain, MCP, OpenEnv, n8n

Big Data: PySpark, Distributed ML, Distributed GPU, Databricks, Azure Data Factory, MLflow, Delta Lake

Databases: PostgreSQL, Qdrant, FalkorDB, Supabase

Cloud & DevOps: Azure AI Foundry, Docker, Kubernetes, FastAPI, REST APIs, WebSockets, GitHub Actions, CI/CD, Prometheus, Grafana

Education

B.E Artificial Intelligence and Machine Learning

CMR Institute of Technology

2022 – 2026

Bengaluru, India